

WASP-52b

R-0461

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_170926 · created 2026-07-19T20:58:46+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458022.8985 +/- 0.0036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.214 +/- 0.063
Transit depth (Rp/Rs)^2	4.58 +/- 2.68 [%]
Orbital Inclination (inc)	89.71 +/- 2.83
Airmass coefficient 1 (a1)	1.16 +/- 0.72
Airmass coefficient 2 (a2)	-0.13 +/- 0.13
Scatter in the residuals of the lightcurve fit is	60.61 %
Best Comparison Star	None
Optimal Aperture	1.34
Optimal Annulus	6.25
Transit Duration (day)	0.0875 +/- 0.0094

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.214 ± 0.063 vs 0.1646 (deviation 0.78σ)
Duration fit vs published	0.0875 d vs 0.0758 d (deviation 1.24σ)
Mid-time O–C	-6.4 min
Depth SNR	3.4
Residual scatter	60.61 %

Light curve

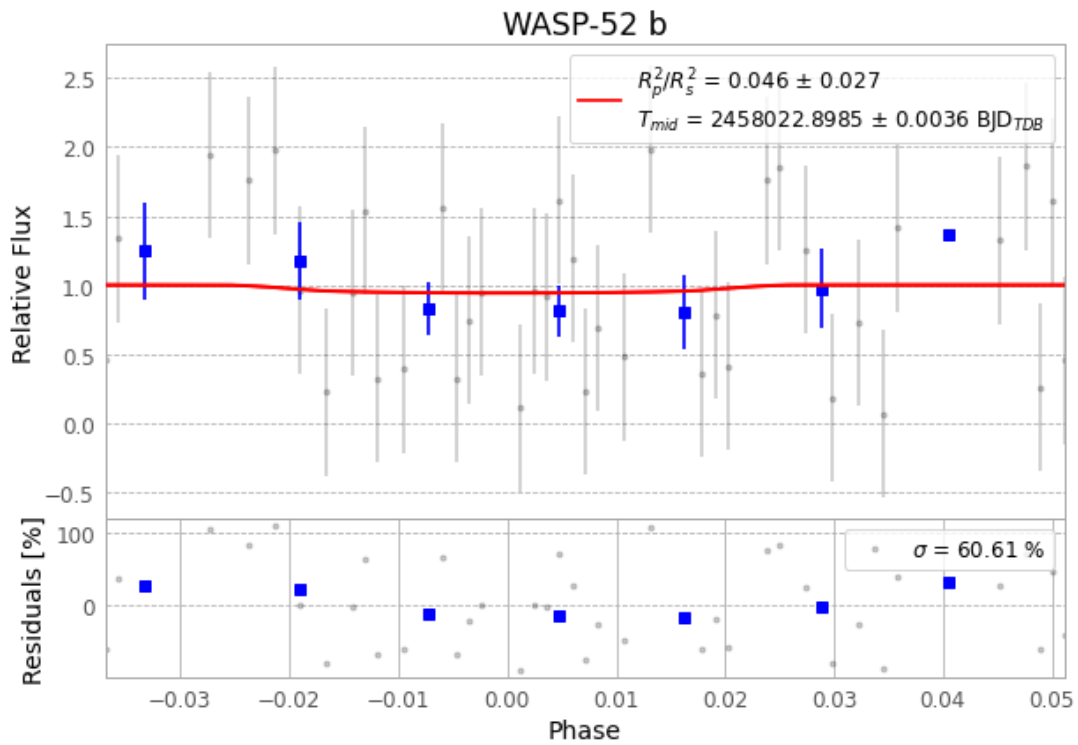


Figure 1 — FinalLightCurve_WASP-52 b_2017-09-26.png (SHA-256 ac70108f1c4dd8e8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [155, 165], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7f76a2754b196bef...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

75 FITS frames (49 MB), WASP-52170926075112.FITS → WASP-52170926113313.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-170926.log (60dc117de59a...), FinalLightCurve_WASP-52 b_2017-09-26.png (ac70108f1c4d...), FinalLightCurve_WASP-52 b_2017-09-26.csv (8d48cc2175d4...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 20:58Z.